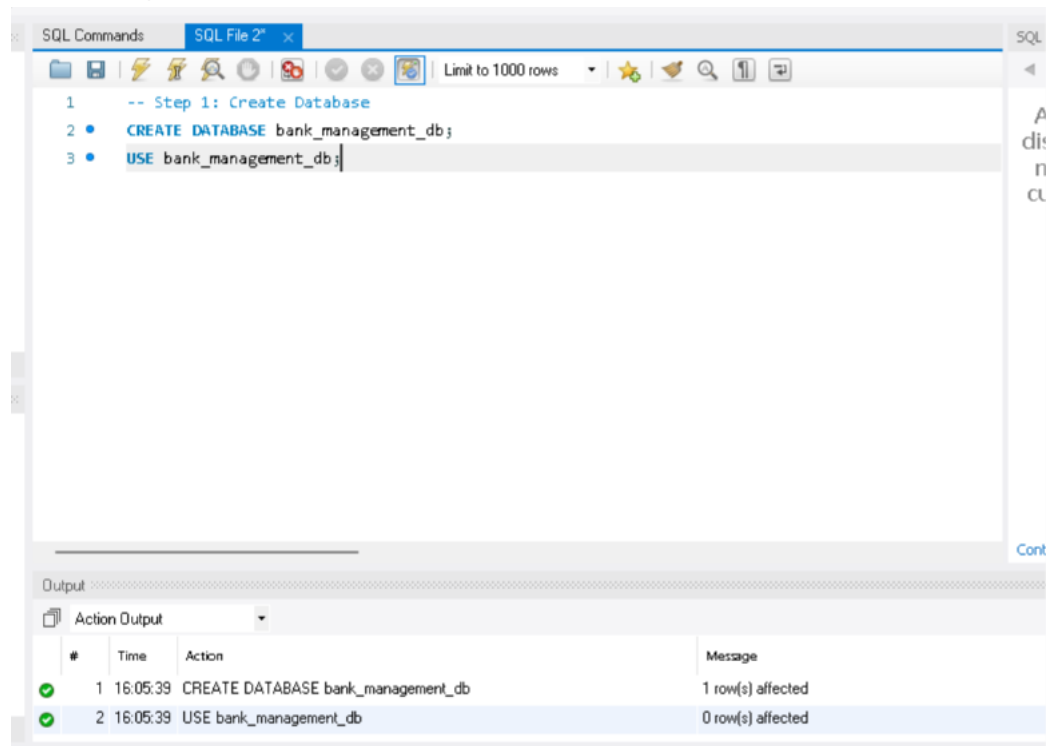


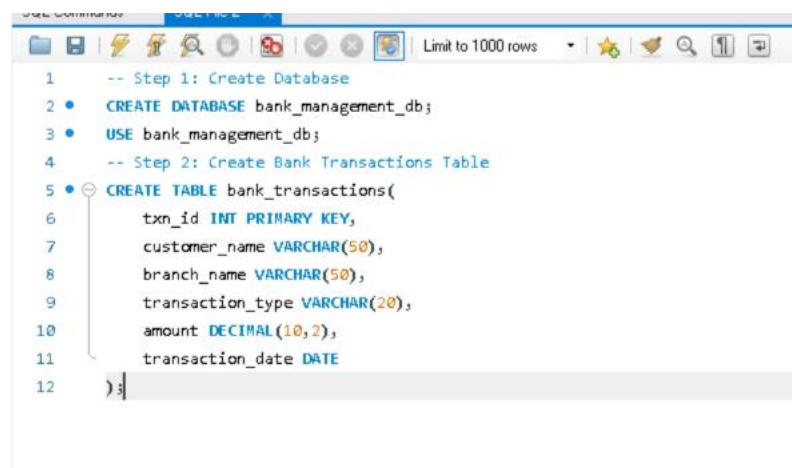
Using DDL, DML, DCL, and TCL commands in your scenarios.

- Using DDL Commands :

1. Creating Data Base:



2. DDL Commands:



```

1  -- Step 1: Create Database
2  • CREATE DATABASE bank_management_db;
3  • USE bank_management_db;
4  -- Step 2: Create Bank Transactions Table
5  • CREATE TABLE bank_transactions(
6      txn_id INT PRIMARY KEY,
7      customer_name VARCHAR(50),
8      branch_name VARCHAR(50),
9      transaction_type VARCHAR(20),
10     amount DECIMAL(10,2),
11     transaction_date DATE
12 );
13 -- Step 3: Add Account Number Column
14 • ALTER TABLE bank_transactions
15   ADD account_no VARCHAR(20);

```

Output

Action Output

#	Time	Action	Message
✓ 18	16:09:20	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected

```

16  -- Step 4: Modify Customer Name Column
17  • ALTER TABLE bank_transactions
18    MODIFY customer_name VARCHAR(100);

```

Output

Action Output

#	Time	Action	Message
✗ 19	16:09:22	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	
✓ 20	16:09:52	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	

```

19  -- Step 5: Rename Transaction Table
20  • RENAME TABLE bank_transactions
21    TO customer_transactions;

```

Output

Action Output

#	Time	Action	Message
✓ 20	16:09:52	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected
✓ 21	16:10:49	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

```

19  -- Step 5: Rename Transaction Table
20  • RENAME TABLE bank_transactions
21    TO customer_transactions;
22  -- Step 6: Truncate Transaction Table
23  • TRUNCATE TABLE customer_transactions;

```

Output

Action Output

#	Time	Action	Message
✓ 21	16:10:49	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
✓ 22	16:11:40	TRUNCATE TABLE customer_transactions	0 row(s) affected

```
24 -- Step 7: Create Backup Table
25 • CREATE TABLE bank_backup(
26     txn_id INT,
27     customer_name VARCHAR(100)
28 );
```

Output

Action Output

#	Time	Action	Message
✓ 22	16:11:40	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓ 23	16:13:02	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR...	0 row(s) affected

```
28 );
29 -- Step 8: Drop Backup Table
30 • DROP TABLE bank_backup;
```

Output

Action Output

#	Time	Action	Message
✓ 23	16:13:02	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR...	0 row(s) affected
✓ 24	16:13:29	DROP TABLE bank_backup	0 row(s) affected

DML COMMANDS:

```
32 -- Step 9: Insert Sample Data
```

```
33 • INSERT INTO customer_transactions
```

```
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
```

```
34 VALUES
```

```
35 (101, 'Hima', 'kurnool', 'Deposit', 5000, '2024-01-05'),
```

```
36 (102, 'Pranitha', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
```

```
37 (103, 'Angela', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
```

```
38 (104, 'Bhanu Priya', 'Vizag', 'Deposit', 8000, '2024-01-10'),
```

```
39 (105, 'Nikshitha', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
```

```
40 (106, 'Manu', 'Vizag', 'Deposit', 15000, '2024-01-12'),
```

```
41 (107, 'Varshika', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
```

```
42 (108, 'Santhvin', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
```

```
43 (109, 'Ravi', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
```

```
44 (110, 'Lahari', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

```
45
```

disat
mar
curr
to

Context I

Output

Action Output

#	Time	Action	Message
25	16:14:29	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_management'
26	16:14:35	INSERT INTO customer_transactions (txn_id, customer_name, branch_name...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```
45 -- Step 10: Insert New Transaction
```

```
46 • INSERT INTO customer_transactions
```

```
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date
```

```
48 VALUES
```

```
49 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
```

Output

Action Output

#	Time	Action	Message
26	16:14:35	INSERT INTO customer_transactions (txn_id, customer_name, branch_name...	10 row(s) affected Records:
27	16:16:36	INSERT INTO customer_transactions (txn_id, customer_name, branch_name...	1 row(s) affected

```
49 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
```

```
50 -- Step 11: Update Transaction Amount
```

```
51 • UPDATE customer_transactions
```

```
52 SET amount = 5000
```

```
53 WHERE txn_id = 105;
```

Output

Action Output

#	Time	Action	Message
27	16:16:36	INSERT INTO customer_transactions (txn_id, customer_name, branch_name...	1 row(s) affected
28	16:17:04	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows match

```
52 SET amount = 5000
```

```
53 WHERE txn_id = 105;
```

```
54 -- Step 12: Delete Transaction
```

```
55 • DELETE FROM customer_transactions
```

```
56 WHERE txn_id = 111;
```

Context Help

Output

Action Output

#	Time	Action	Message
28	16:17:04	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
29	16:17:39	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected

manu
curren
togg

```
54 -- Step 12: Delete Transaction
55 • DELETE FROM customer_transactions
56 WHERE txn_id = 111;
57 -- Step 13: Display All Transactions
58 • SELECT *
59 FROM customer_transactions;
60
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
▶	101	Hima	kurnool	Deposit	5000.00	2024-01-05	NULL
	102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
	103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
	104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
	105	Nikshitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

transactions 1 x

Apply Revert

Context Help

Output

Action Output

#	Time	Action	Message
✓ 29	16:17:39	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 30	16:18:27	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned

/ di r ci

```
56 WHERE txn_id = 111;
57 -- Step 13: Display All Transactions
58 • SELECT *
59 FROM customer_transactions;
60 -- Step 14: Display Deposit Transactions
61 • SELECT *
62 FROM customer_transactions
63 WHERE transaction_type = 'Deposit';
64
```

Limit to 1000 rows

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

Result Grid

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
▶	101	Hima	kurnool	Deposit	5000.00	2024-01-05	NULL
	103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
	104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
	106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
	108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL

customer_transactions 2 x

Apply Revert

Con

Output

Action Output

#	Time	Action	Message
✓ 30	16:18:27	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 31	16:19:19	SELECT * FROM customer_transactions WHERE transaction_type = 'D...	6 row(s) returned

```

61 • SELECT *
62 FROM customer_transactions
63 WHERE transaction_type = 'Deposit';
64 -- Step 15: Sort Transactions by Amount
65 • SELECT *
66 FROM customer_transactions
67 ORDER BY amount DESC;
68

```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL

customer_transactions 3 x Apply

Output

#	Time	Action	Message
31	16:19:19	SELECT * FROM customer_transactions WHERE transaction_type = 'D...	6 row(s) returned
32	16:20:24	SELECT * FROM customer_transactions ORDER BY amount DESC LIM...	10 row(s) returned

DCL COMMANDS:

```

67 ORDER BY amount DESC;
68 -- Create Auditor1 User
69 • CREATE USER 'Auditor1'@'localhost'
70 IDENTIFIED BY '1234';
71 -- Create BranchManager User
72 • CREATE USER 'BranchManager'@'localhost'
73 IDENTIFIED BY '1234';
74

```

Output

#	Time	Action	Message
32	16:20:24	SELECT * FROM customer_transactions ORDER BY amount DESC LIM...	10 row(s) returned
33	16:21:54	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected

```

74 -- Step 16: Grant SELECT Permission
75 • GRANT SELECT
76 ON customer_transactions
77 TO 'Auditor1'@'localhost';

```

Output

Action Output

#	Time	Action	Message
34	19:45:50	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_mana
35	19:45:52	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected

```

77 TO 'Auditor1'@'localhost';
78 -- Step 17: Grant All Privileges
79 • GRANT ALL
80 ON customer_transactions
81 TO 'BranchManager'@'localhost';

```

Conte

Output

Action Output

#	Time	Action	Message
33	16:21:54	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
34	19:45:50	CREATE DATABASE bank_management_db	Error Code: 1007. Can't create database 'bank_management_
35	19:45:52	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected

Commands

```

63 SELECT transaction_type = 'Deposit';
64 -- Step 15: Sort Transactions by Amount
65 • SELECT *
66 FROM customer_transactions
67 ORDER BY amount DESC;
68 -- Create Auditor1 User
69 • CREATE USER 'Auditor1'@'localhost'
70 IDENTIFIED BY '1234';
71
72 -- Create BranchManager User
73 • CREATE USER 'BranchManager'@'localhost'
74 IDENTIFIED BY '1234';
75 -- Step 16: Grant SELECT Permission
76 • GRANT SELECT
77 ON customer_transactions
78 TO 'Auditor1'@'localhost';
79 -- Step 17: Grant All Privileges
80 • GRANT ALL
81 ON customer_transactions
82 TO 'BranchManager'@'localhost';

```

Output

Action Output

#	Time	Action	Message
17	22:34:28	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected

Commands

```

67 ORDER BY amount DESC;
68 -- Create Auditor1 User
69 • CREATE USER 'Auditor1'@'localhost'
70 IDENTIFIED BY '1234';
71
72 -- Create BranchManager User
73 • CREATE USER 'BranchManager'@'localhost'
74 IDENTIFIED BY '1234';
75 -- Step 16: Grant SELECT Permission
76 • GRANT SELECT
77 ON customer_transactions
78 TO 'Auditor1'@'localhost';
79 -- Step 17: Grant All Privileges
80 • GRANT ALL
81 ON customer_transactions
82 TO 'BranchManager'@'localhost';
83 -- Step 18: Revoke SELECT Permission
84 • REVOKE SELECT
85 ON customer_transactions
86 FROM 'Auditor1'@'localhost';

```

Output

Action Output

#	Time	Action	Message
18	22:36:08	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected

```

84 • REVOKE SELECT
85 ON customer_transactions
86 FROM 'Auditor1'@'localhost';
87 -- Step 19: Revoke All Privileges
88 • REVOKE ALL PRIVILEGES
89 ON customer_transactions
90 FROM 'BranchManager'@'localhost';
91

```

Output

#	Time	Action	Message
19	22:36:32	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	22:42:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected

TCL COMMANDS:

```

86 FROM 'Auditor1'@'localhost';
87 -- Step 19: Revoke All Privileges
88 • REVOKE ALL PRIVILEGES
89 ON customer_transactions
90 FROM 'BranchManager'@'localhost';
91 -- Step 20: Commit Transaction
92 • UPDATE customer_transactions
93 SET amount = 6000
94 WHERE txn_id = 101;
95
96 • COMMIT;
97

```

Output

#	Time	Action	Message
20	22:38:33	CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	22:42:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	22:44:29	COMMIT	0 row(s) affected

```

88 • REVOKE ALL PRIVILEGES
89 ON customer_transactions
90 FROM 'BranchManager'@'localhost';
91 -- Step 20: Commit Transaction
92 • UPDATE customer_transactions
93 SET amount = 6000
94 WHERE txn_id = 101;
95
96 • COMMIT;
97 -- Step 21: Create Savepoint
98 • SAVEPOINT Before_Update;
99

```

Output

#	Time	Action	Message
21	22:39:08	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
22	22:39:30	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
23	22:39:55	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected
24	22:42:37	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
25	22:44:29	COMMIT	0 row(s) affected
26	22:45:17	SAVEPOINT Before_Update	0 row(s) affected


```

96 • COMMIT;
97 -- Step 21: Create Savepoint
98 • START TRANSACTION;
99
100 • SAVEPOINT Before_Update;
101 -- Step 22: Rollback to Savepoint
102 • UPDATE customer_transactions
103   SET amount = 99999
104   WHERE txn_id = 102;
105

```

Output			
Action Output			
#	Time	Action	Message
✓ 26	22:45:17	SAVEPOINT Before_Update	0 row(s) affected
✗ 27	22:45:43	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
✓ 28	22:45:56	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✗ 29	22:46:04	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
✓ 30	22:47:26	SAVEPOINT Before_Update	0 row(s) affected
✓ 31	22:47:48	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0

Output			
Action Output			
#	Time	Action	Message
109		-- Reset Transaction State	
110		ROLLBACK;	
111			
112		SET autocommit = 1;	
113		-- Step 23: Complete TCL Example	
114		START TRANSACTION;	
115			
116		UPDATE customer_transactions	
117		SET amount = 7000	
118		WHERE txn_id = 101;	
119			
120		SAVEPOINT SP1;	
121			
122		UPDATE customer_transactions	
123		SET amount = 9000	
124		WHERE txn_id = 102;	
125			
126		ROLLBACK TO SAVEPOINT SP1;	
127			
128		COMMIT;	
129			
✓ 33	22:50:24	COMMIT	0 row(s) affected
✓ 34	22:50:40	SAVEPOINT Before_Update	0 row(s) affected
✗ 35	22:51:04	ROLLBACK TO Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
✓ 36	22:53:20	SET autocommit = 1	0 row(s) affected
✗ 37	22:53:48	ROLLBACK TO SAVEPOINT Before_Update	Error Code: 1305. SAVEPOINT Before_Update does not exist
✓ 38	22:55:24	COMMIT	0 row(s) affected

